Community and Ecosystem Ecology

Lesson 3



Community and Ecosystem Ecology

Community Ecology

- Community Structures
- Community Dynamics
- Species Interaction

Ecosystem Ecology

- Energy Flow Through Ecosystems

Learning objectives

At the end of the lesson, students should be able to:

1. Describe **community structure** and its components.
2. Explain factors that influence **community dynamics.**
3. Distinguish between **succession types.**
4. Identify **different types of interactions** among organisms.
5. Illustrate **the flow of energy and matte**r in ecosystems.

Community Structure

Community structure refers to the **organization and composition of species** in a community.

Species Richness

- **Species richness** is the **number of different species** present in each area or ecosystem.
- It focuses **only on the count**, not how many individuals each species has.
- Higher richness = more variety of species.

Example:
If a forest has:

- 5 species of trees → Species richness = **5**

Importance:

- Species richness tells us how diverse an ecosystem is.
- Helps assess ecosystem health (more species often = more stable ecosystem).

Relative Species Abundance

- **Relative species abundance** refers to **how many individuals each species has compared to the total number of individuals in the ecosystem.**

Formula:

- Relative *Abundance* $= \frac{Number\ of\ individuals\ of\ a\ species}{Total\ number\ of\ individuals\ of\ all\ species}$

- Shows **distribution or evenness** of species.
- Tells whether a species is **common or rare**.
- Tells us how many individuals of each species are present.

Example:

In a sample with 100 organisms:

- Species A = 70 individuals → 70%
- Species B = 20 individuals → 20%
- Species C = 10 individuals → 10%

- **Species richness = 3** (there are 3 different species).

- **Species abundance** is uneven:
  - Species A = 70%
  - Species B = 20%
  - Species C = 10%

Although the sample has a species richness of 3, the uneven abundance shows low species evenness, meaning one species is much more common than the others.

This suggests **lower overall biodiversity** compared to a community where all three species have similar abundances.

DOMINANT SPECIES

- Species with the **highest abundance or biomass.**
- **Strong influence** on the environment and other species.
  - Dominant species affect:
    - Resource availability (light, water, nutrients)
    - Habitat structure
    - Other species’ survival and distribution

How Dominant Species Shape Communities

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1. Resource Control

- Dominant species often monopolize important resources:
  - Light (tall trees block sunlight)
  - Water (deep-rooted plants access more water)
  - Nutrients (fast-growing plants absorb more nutrients)

Result: Limits the growth of other species

2. Habitat Formation

- They create **physical environments** where other organisms live.

Example:

- Trees provide shelter, nesting areas, and shade.
- Corals create reefs where fish live.

How Dominant Species Shape Communities

3. Food Web Influence

- They often serve as a **primary food source.**
- Affect energy flow in the ecosystem.

Example:

- Grass → eaten by herbivores → supports predators

4. Community Stability

- Their presence helps maintain **balance.**
- Removing them may cause major changes.

Examples of Dominant Species

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	Dominant Species	Role
Forest	Oak or pine trees	Control light, provide habitat.
Grassland	Grasses	Main food source for herbivores.
Coral reef	Corals	Form physical structure of ecosystem.
Marine	Phytoplankton	Base of food chain

FOUNDATIO N SPECIES

- A **foundation species** is a species that **creates, shapes, or maintains** the **habitat** for other organisms.

Key Characteristics:

- Usually **abundant and dominant** in the ecosystem.
- **Physically shapes the environment.**
- Creates **habitat or shelter** for other species.
- Influences **biodiversity indirectly.**

Examples

1. Trees in a Forest

- Example: Oak or pine trees
- Provide:
  - Shelter for animals
  - Shade controlling temperature
  - Nutrients through leaf litter
- Without them → forest ecosystem would collapse

2. Coral in Coral Reefs

- Coral builds the **reef structure**
- Provides habitat for:
  - Fish
  - Invertebrates
  - Algae

Examples

3. Seagrass in Marine Ecosystems

- Provides nursery grounds for fish.
- Stabilizes ocean floor sediments.
- Supports marine food webs.

4. Mangrove Trees

- Serve as breeding grounds for aquatic species.
- Found in coastal areas.
- Protect shorelines from erosion.

KEYSTONE SPECIES

- A **keystone species** is a species that has a **very large impact on its ecosystem relative to its abundance**.
- Even if it is not the most common species, its presence helps maintain the balance of the ecosystem.

Importance

- Controls populations of other species.
- Maintains biodiversity.
- Prevents one species from dominating the ecosystem.
- Supports ecosystem stability.

Examples:

1. **Sea Otters**
   - Eat sea urchins.
   - Prevent sea urchins from destroying kelp forests.

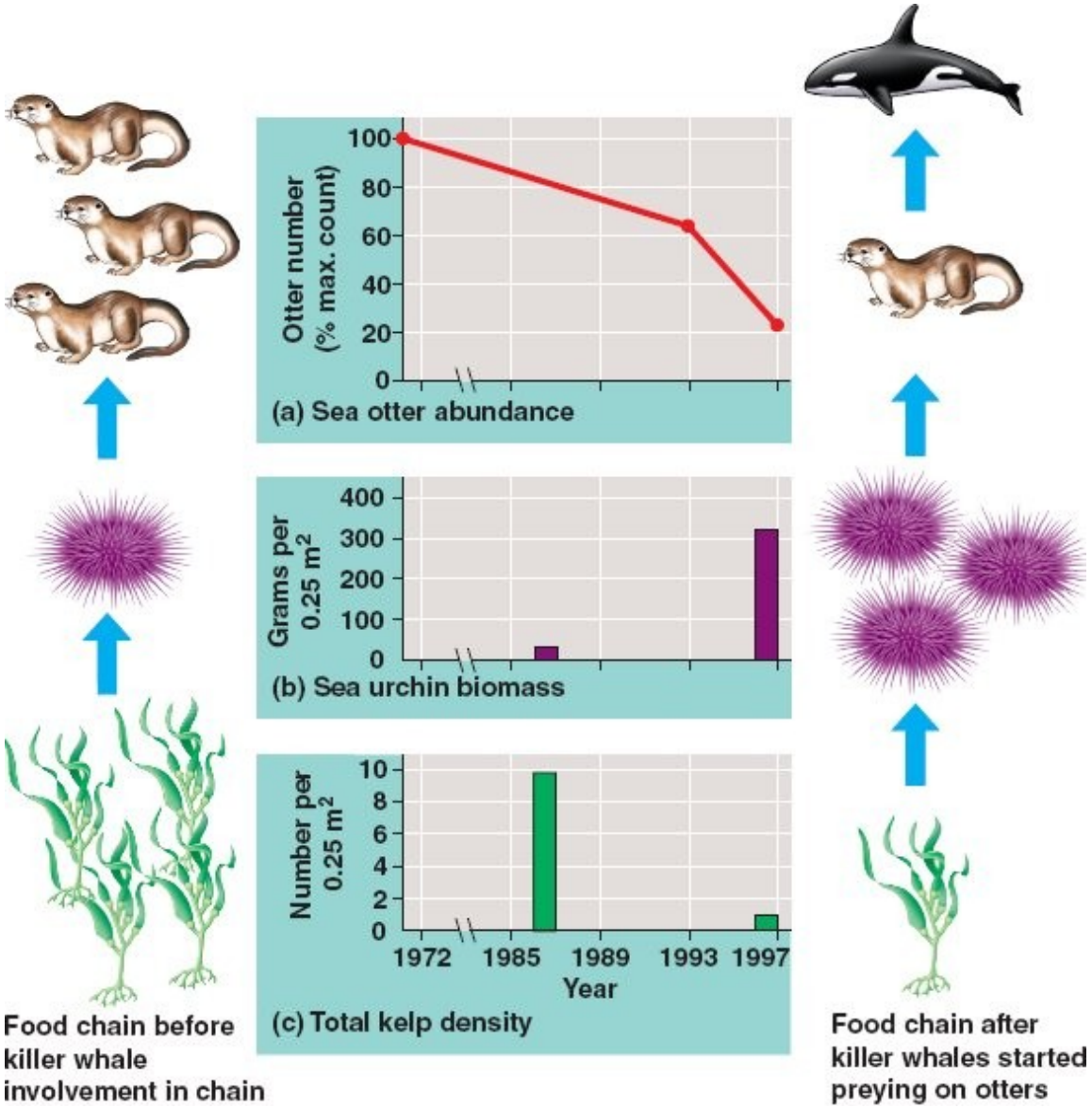
2. **Wolves (Yellowstone National Park)**
   - Control deer and elk populations.
   - Allow vegetation and other animal populations to recover.

3. **Bees**
   - Pollinate many flowering plants and crops.
   - Support plant reproduction and food webs.

4. **Starfish (Pisaster)**
   - Control mussel populations.
   - Allow many other species to coexist.

What happens if a keystone species is removed?

- One species may become overly abundant.
- Biodiversity decreases.
- Food webs become disrupted.
- The ecosystem may partially or completely collapse.


Otter number (% max. count)
100
80
60
40
20
0
(a) Sea otter abundance
Grams per 0.25 m²
400
300
200
100
0
(b) Sea urchin biomass
Number per 0.25 m²
10
8
6
4
2
0
1972
1985
1989
1993
1997
Year
(c) Total kelp density
Food chain before killer whale involvement in chain
Food chain after killer whales started preying on otters



DID YOU KNOW?

Wolves are an essential keystone species.



In Yellowstone, the return of the wolf helped correct the destructive over-population of elk and coyotes, allowing plants and animals to thrive.

With fewer coyotes, there are more small animals like rabbits and voles that other predators and birds of prey rely on for food.

With fewer elk, the aspens and willows have grown back, revitalizing Yellowstone's landscape, and inviting the return of another ecosystem engineer - the beaver.

Summary of Ecological terms

- **Species Richness**
  - The **number of different species** present in an ecosystem.
- **Relative Species Abundance**
  - The **proportion or percentage of individuals** belonging to each species compared to the total population.
- **Dominant Species**
  - The species with the **highest abundance or biomass**, exerting the greatest influence on the community due to its large numbers.
  - Most numerous but not always ecosystem-shaping.
- **Foundation Species**
  - A species that **creates, structures, or maintains a habitat** that supports many other organisms.
  - Creates or defines habitat
    - Example: Corals in coral reefs, mangroves in coastal ecosystems.
- **Keystone Species**
  - A species whose **impact on the ecosystem is much greater than its abundance**. Removing it can cause major changes in ecosystem structure and biodiversity.
  - Has a large impact despite low abundance.
    - Example: Wolves, sea otters, starfish.

Community Dynamics

Community dynamics refers to **changes in community structure over time**.

Ecological Succession

- Ecological succession is the **process** by which **ecosystems recover, change, and reach stability** over time after a **disturbance.**

Disturbance

- Any event that **disrupts an ecosystem** (natural or human-made).
  - Examples: wildfire, typhoon, deforestation.
- Disturbances often **restart succession**.

Ecological Succession

Pioneer species

- **Pioneer species** (like moss and lichens) are the first to grow.
- These organisms help form soil, allowing larger plants to grow later.

Climax Community

- The **final, stable stage** of succession.
- The ecosystem becomes balanced, with mature plants and animals.

Types of Ecological Succession

1. Primary Succession

- **Primary succession** is the process by which a community of organisms develops in an area where **no soil and no life previously existed**.

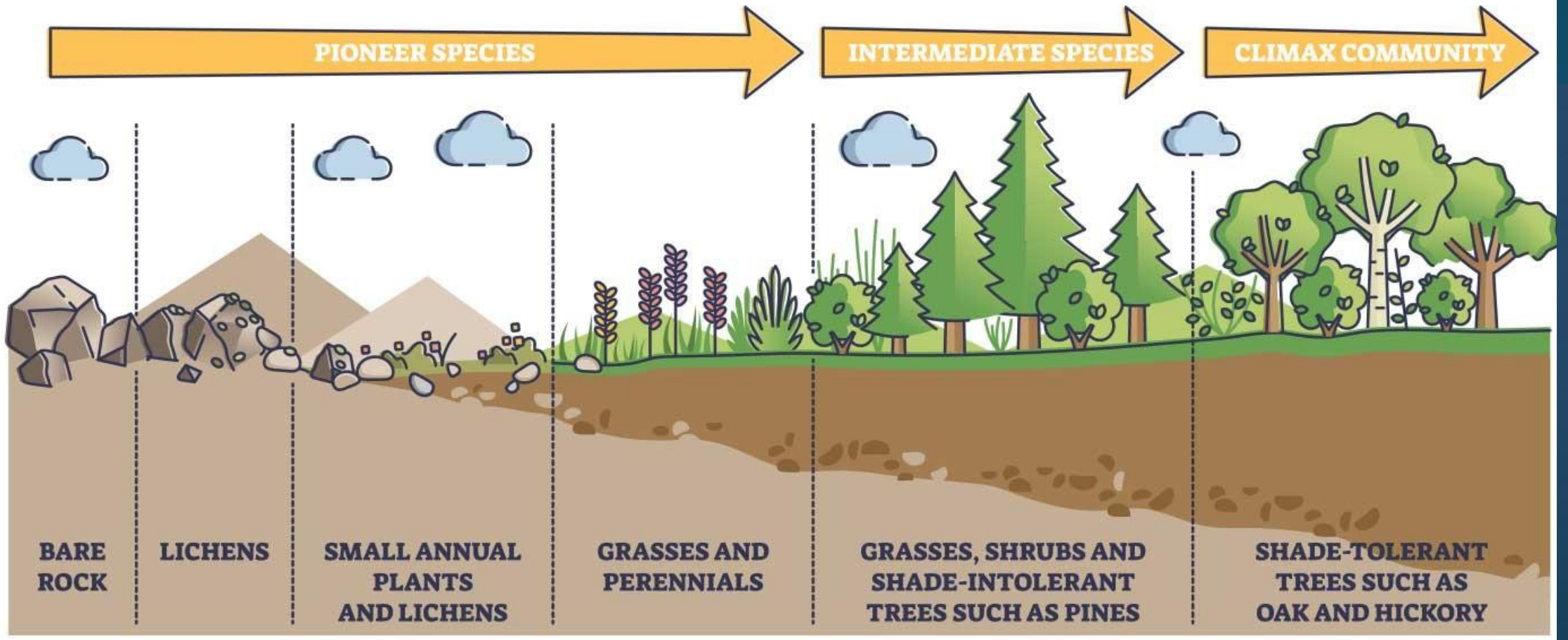
Examples

- New volcanic islands
- Land exposed after glaciers melt
- Areas covered by lava flows

How it Happens?

1. **Bare rock is exposed** (e.g., after a volcanic eruption or retreating glacier).
2. **Pioneer species,** such as lichens and mosses, colonize the area.
3. These organisms break down rock and help form **soil**.
4. Grasses, shrubs, and small plants begin to grow.
5. Trees and larger organisms establish themselves.
6. A stable **climax community** eventually develops.

PRIMARY SUCCESSION



Types of Ecological Succession

2. Secondary Succession

- **Secondary succession** is the process by which an ecosystem recovers after a disturbance that destroys a community **but leaves the soil intact**.

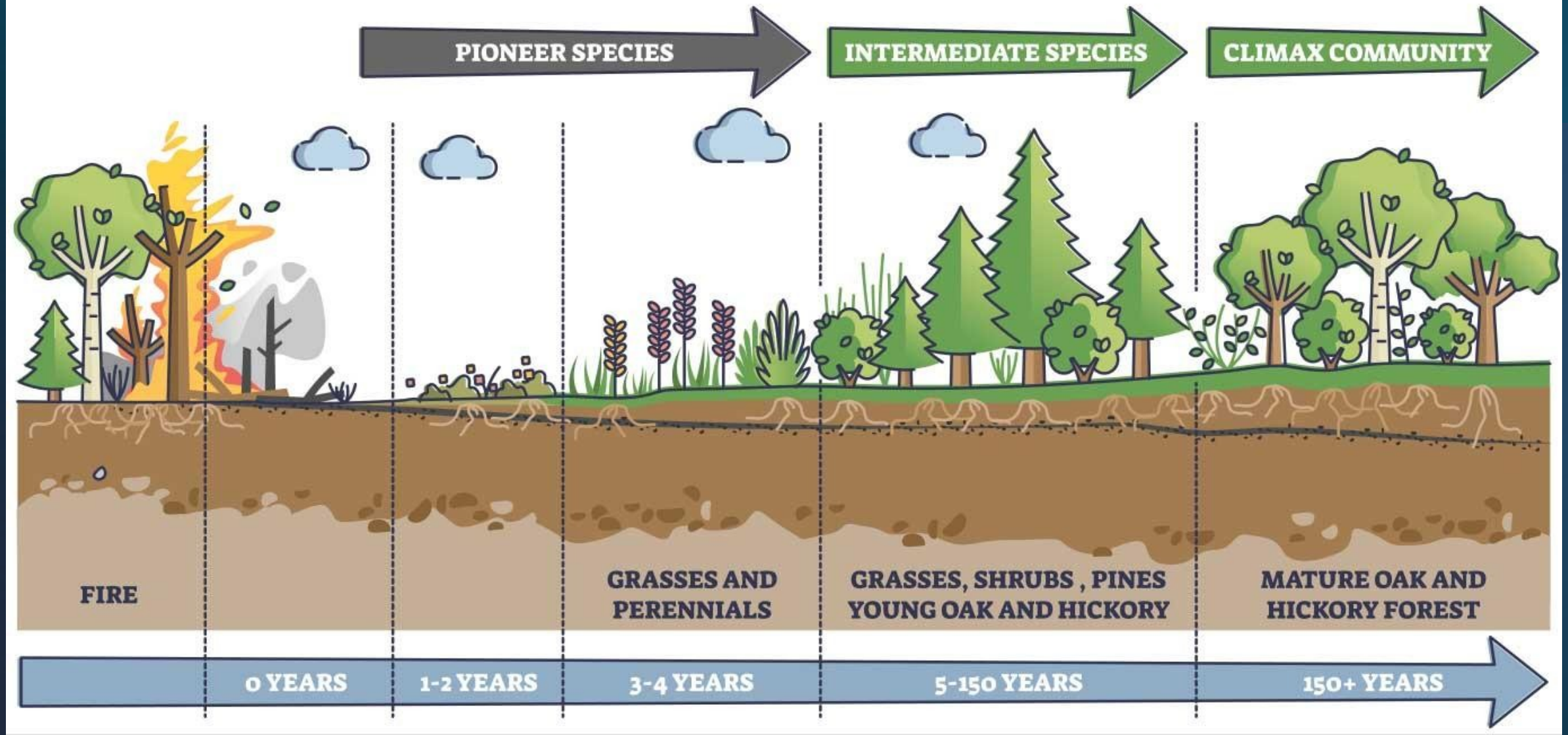
Examples

- Forest regrowth after a wildfire.
- Abandoned farmland returning to forest.
- Recovery after a flood or hurricane.

How it Happens

1. A disturbance occurs (fire, flood, farming, logging, hurricane).
2. Soil remains in place.
3. Grasses and small plants grow first.
4. Shrubs and young trees appear.
5. Larger trees and animals return.
6. A stable ecosystem is re-established.

SECONDARY SUCCESSION



Primary versus Secondary Succession

	Primary Succession	**Secondary Succession**
Soil present?	No	Yes
Starts with	Bare rock	Existing soil
Pioneer species	Lichens and mosses	Grasses and herbs
Speed	Slow	Faster
Example	New volcanic island	Forest after a fire

Species Interactions

Species interactions refer to the **relationships** between different species that coexist in the same ecosystem.

These interactions **affect** the **survival, growth**, and **population** of the species involved.

Types of Species Interactions

A. Competition

- Intraspecific competition
- Interspecific competition

B. Predation

C. Herbivory

C. Symbiosis (Closed Relationship)

- Mutualism
- Commensalism
- Parasitism

A. COMPETITION (-/-)

- **Competition** happens when organisms **fight for the same limited resources** (like food, space, water, or mates).
- **Both** species are **negatively affected.**

1. Intraspecific competition occurs between members of the **SAME** species

- Two dogs fighting for food.
- Trees of the same species competing for sunlight.
- Deer competing for mates

2. Interspecific competition occurs between **DIFFERENT** species.

- Lions and hyenas hunting the same prey.
- Plants of different species competing for water .
- Birds and squirrels competing for seeds.

Cont.

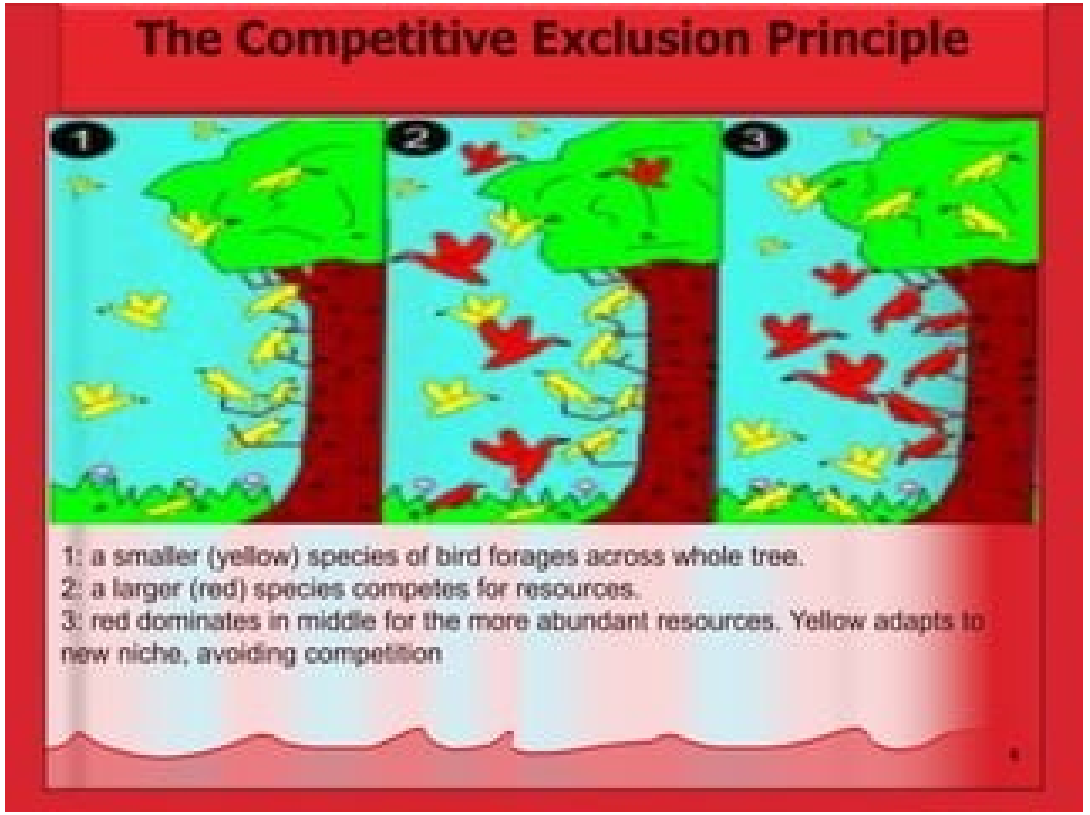
Competitive exclusion principle

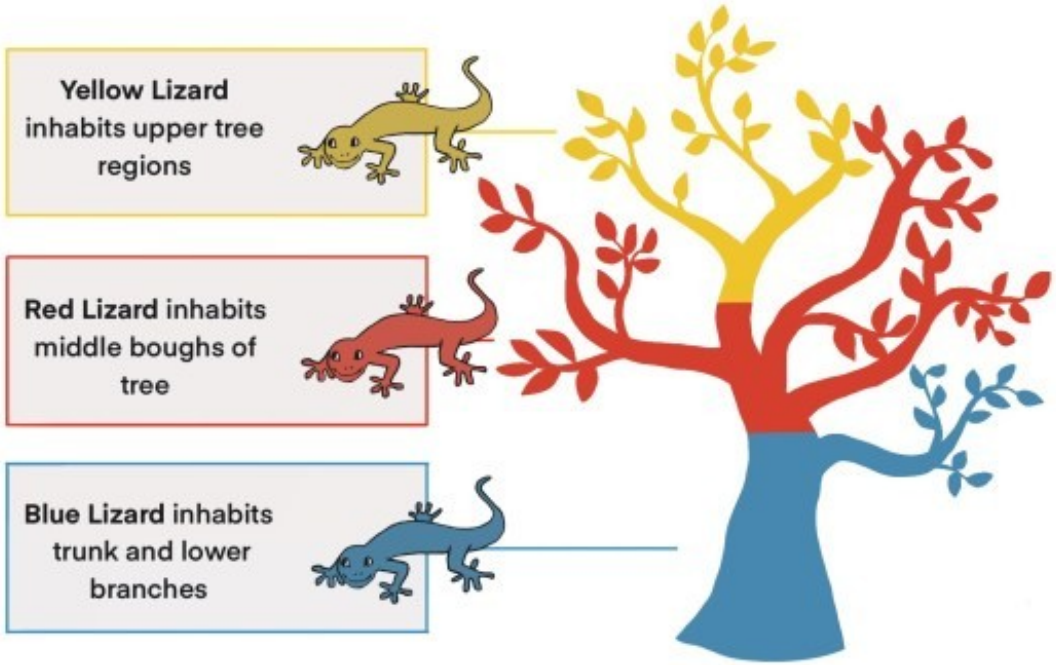
- **Competitive exclusion principle** states that **two species cannot occupy** the **same niche** in a habitat indefinitely.
- Species competing for the same resources **cannot coexist forever.**
- One species will eventually outcompete or replace the other.

This exclusion may be avoided.

Resource partitioning

- **Resource partitioning** is when species divide resources **to reduce competition.**
- Each species uses the resources in a **different way, time, or place**.
- This allows species to **coexist in the same habitat** without outcompeting each other.





Resource partitioning

B. Predation (+/-)

- One species (predator) **kills and eats** another species (prey).
- **Predator** benefits; **prey** is harmed.

- Examples
  - Lion → eats zebra
  - Snake → eats mouse
  - Bird → eats insect



Defense Mechanisms Against Predation

Defense Mechanism	Description	Example	How It Helps Survival
Camouflage	Blends with environment.	Stick insect, chameleon	Avoids being seen by predators.
Warning Coloration	Bright colors warn predators of danger.	Poison dart frog	Predators avoid eating them.
Mimicry	Harmless species copies harmful one.	King snake (looks venomous)	Tricks predators into staying away.
Physical Defenses	Body parts protect the organism.	Turtle (shell), porcupine(spikes), plants (thorn)	Makes it hard to attack or eat.
Speed/Escape	Moves quickly to escape.	Deer running fast	Helps avoid being caught.
Chemical Defenses	Produces toxins or bad smells.	Skunk spraying foul odor	Repels or harms predators.
Behavioral Defenses	Actions to avoid danger.	Playing dead (opossum)	Confuses or discourages predators.

C. Herbivory (+/-)

- An animal feeds on **plants** or **algae.**
- **Herbivore** benefits; the **plant** is harmed.
- Example: Cow grazing on grass.



D. Symbiosis

Mutualism (+/+)

Mutualism (+/+)

- **Both** organisms **benefit.**

Example: Bees get nectar from flowers while helping pollinate them.



D. Symbiosis

Commensalism (+/0)

Commensalism

- One **benefits,** the other is **unaffected** (neither helped nor harmed).

- Example:
  - Birds nesting in trees



D. Symbiosis

Parasitism (+/-)

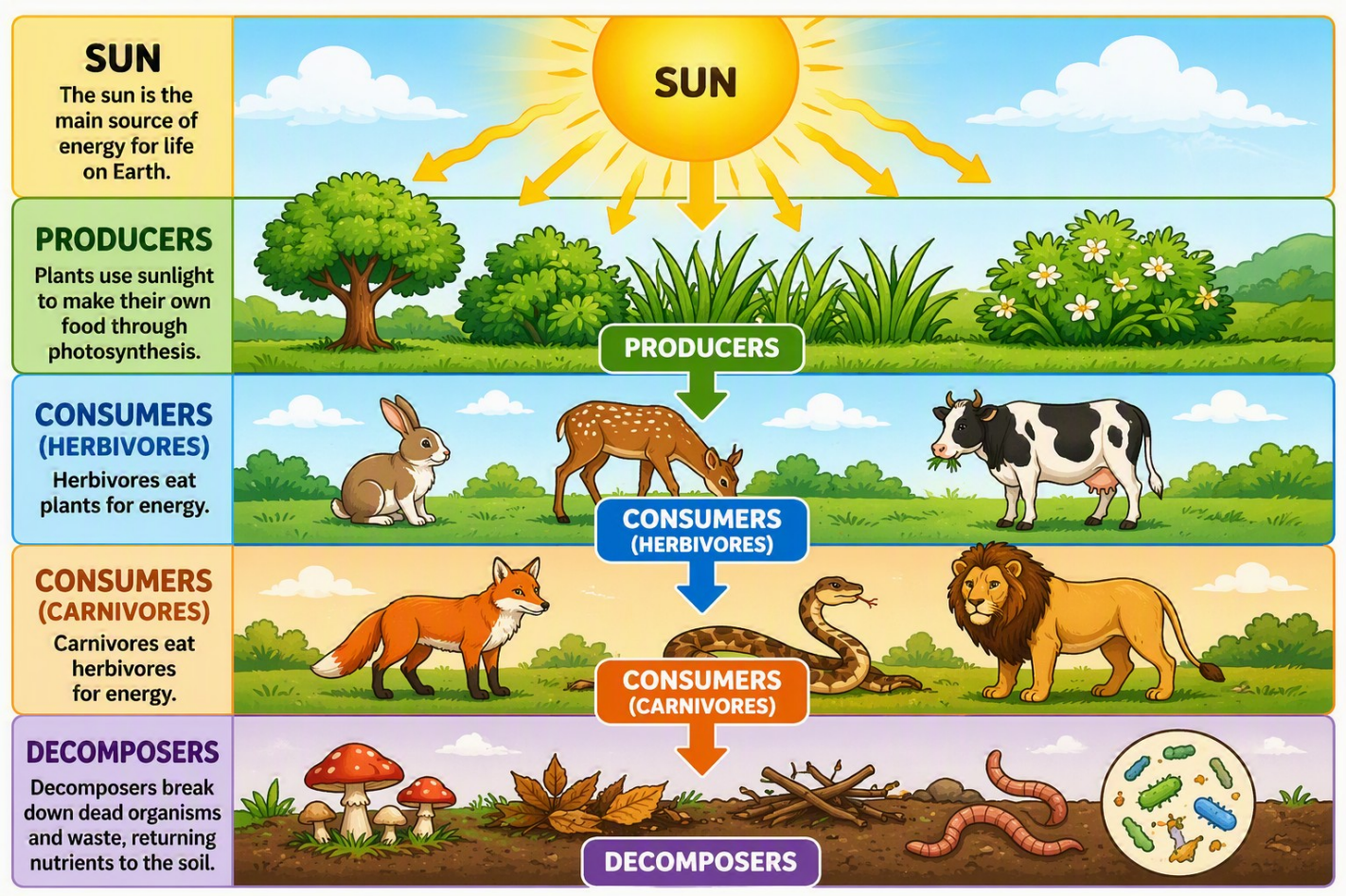
Parasitism

- A parasite lives on or in a host and obtains nutrients from it.
- **Parasite** benefits; the **host** is harmed.
- Example: Ticks on dogs



Interaction	Effect on Species 1	Effect on Species 2	Example
Competition	-	-	Lions and hyenas
Predation	+	-	Snake and mouse
Herbivory	+	-	Cow and grass
Mutualism	+	+	Bee and flower
Commensalism	+	0	Barnacle and whale

Energy Flow in the Ecosystem



Energy comes mainly from the **Sun**.

Passed through organisms through **food chains** and **food webs**.

Food Chains

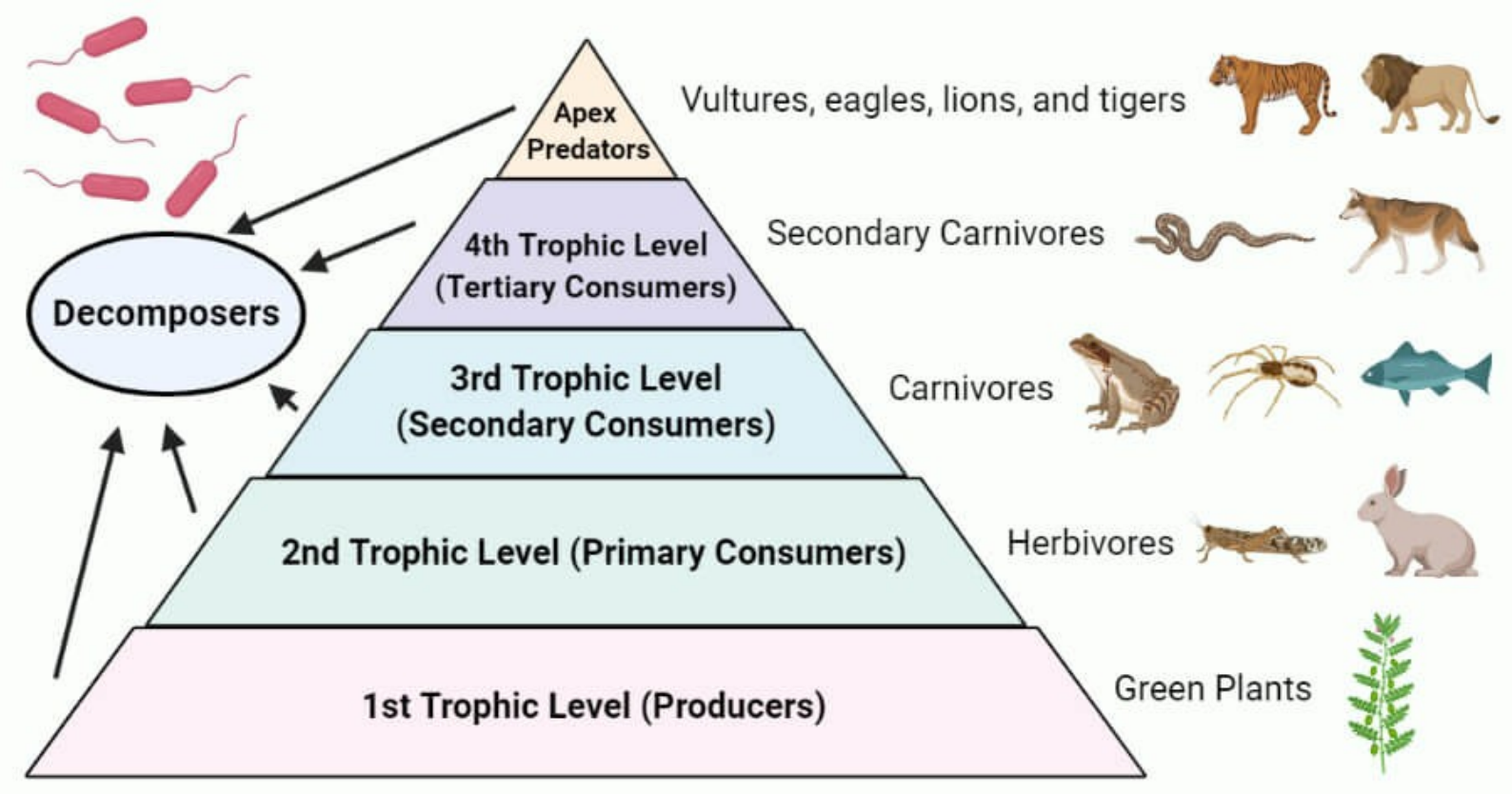
- A **food chain** is a linear sequence of organisms through which **nutrients and energy** pass as one organism eats another.
- There is a single path through a food chain.

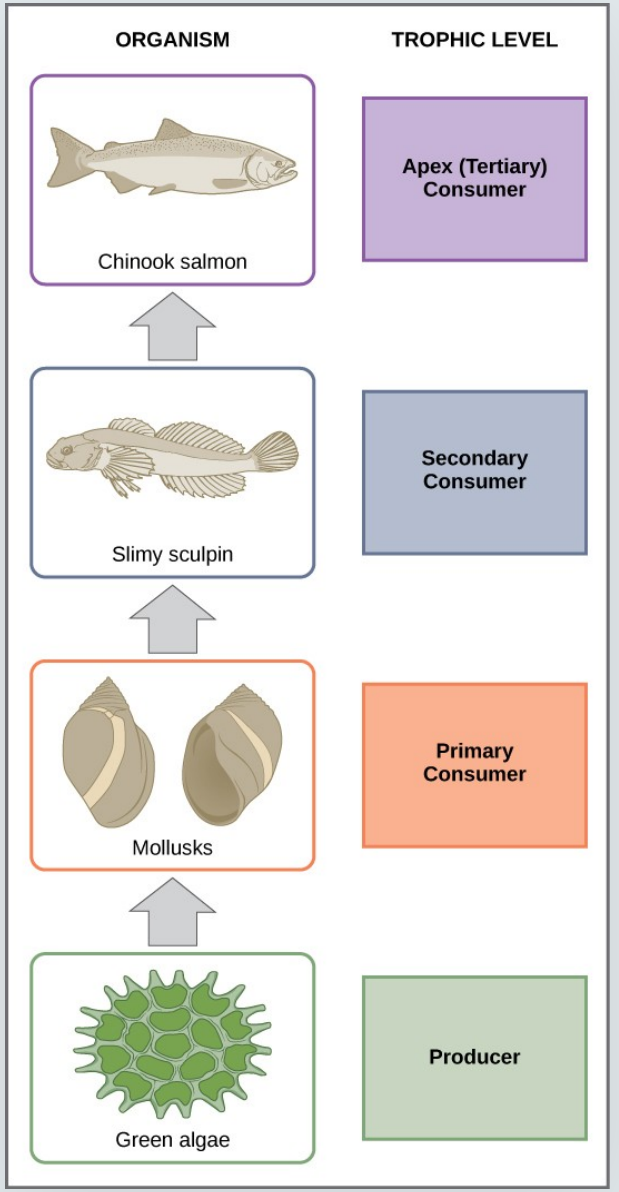


Trophic level

- Each organism in a food chain occupies a specific **trophic level** (energy level), **its position** in the food chain.
- **Producers (1st Level)**
  - Plants and algae
  - Make their own food using sunlight (**photosynthesis**)
- **Primary Consumers (2nd Level)**
  - Herbivores (eat plants)
  - Examples: rabbit, deer, grasshopper
- **Secondary Consumers (3rd Level)**
  - Carnivores that eat herbivores
  - Examples: snake, fox
- **Tertiary Consumers (4th Level)**
  - Top predators
  - Examples: eagle, lion
- **Higher Trophic Levels:** Some ecosystems have **quaternary consumers** and other top predators
- **Decomposers** break down dead and decaying organisms from any trophic level.

Trophic levels of a food chain





10% Energy Rule

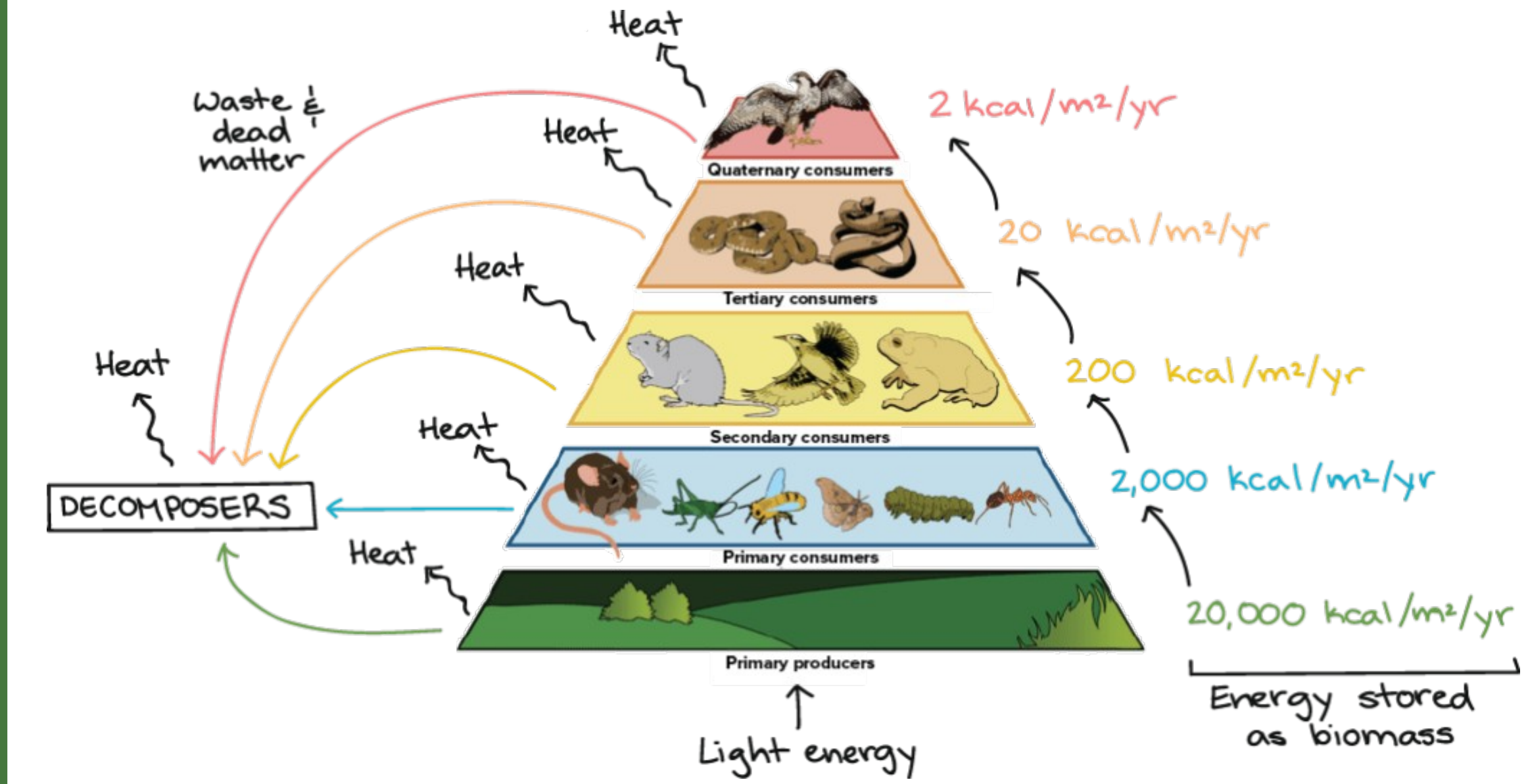
- The **10% energy rule** states that **only about 10% of the energy at one trophic level is transferred to the next trophic level**.
- The remaining **90% is lost** as heat, movement, growth, and other life processes.

Example

- **Producers (Grass)** = 10,000 units of energy
- **Primary Consumers (Rabbit)** = 1,000 units
- **Secondary Consumers (Snake)** = 100 units
- **Tertiary Consumers (Hawk)** = 10 units

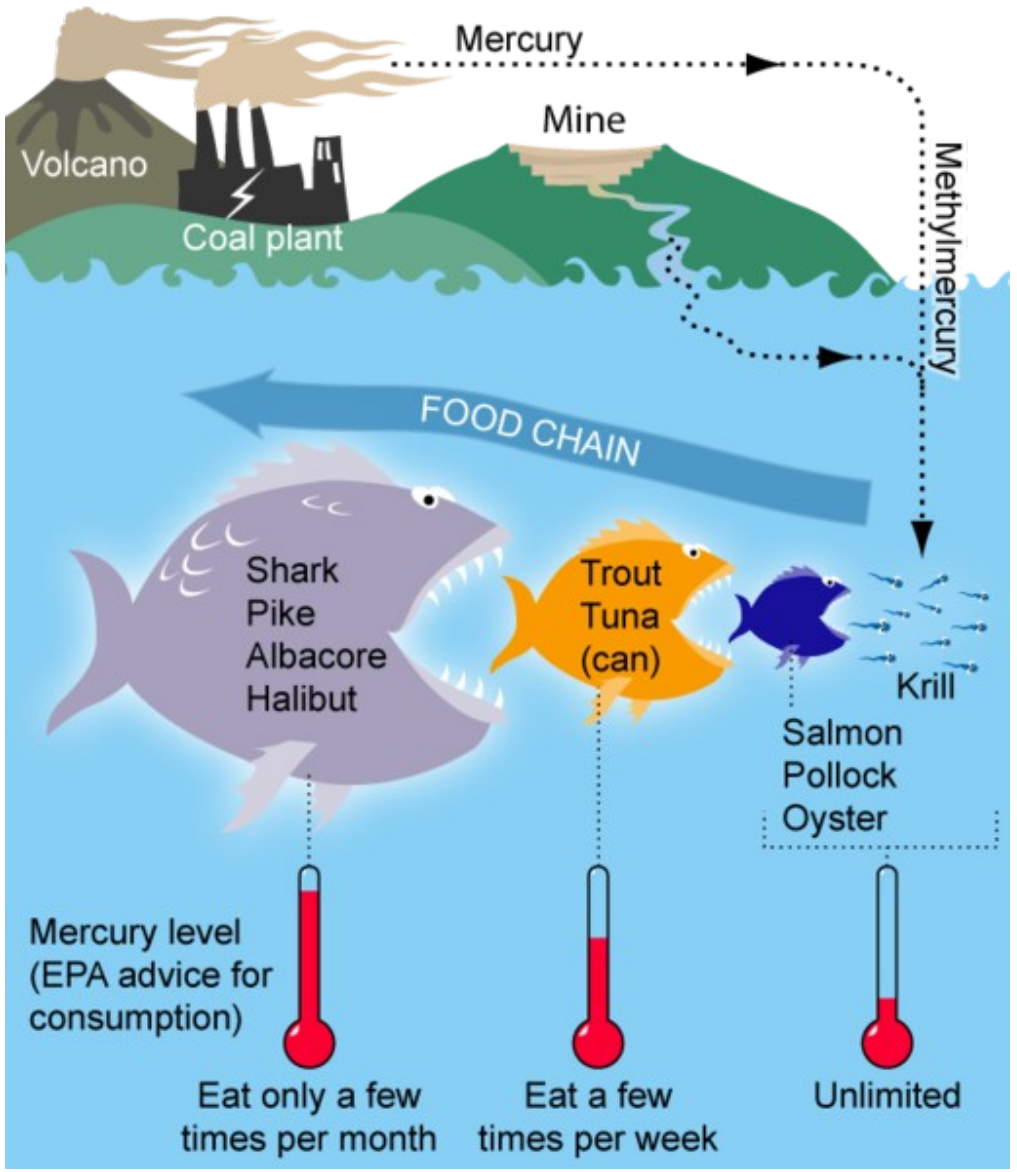
Why is it important?

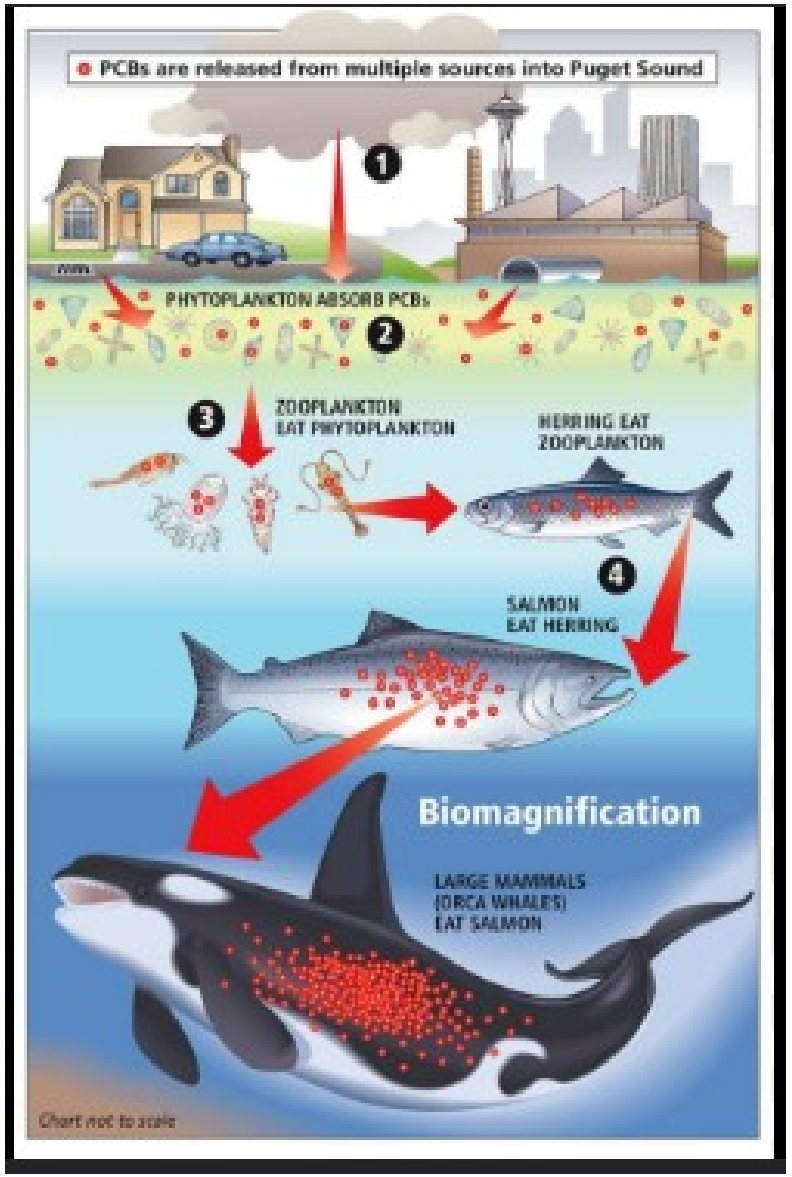
- Explains why food chains are usually short.
- Shows why there are fewer top predators than producers.
- Demonstrates the flow and loss of energy in ecosystems.


Heat
Waste & dead matter
Heat
Quaternary consumers
2 kcal/m²/yr
Heat
Tertiary consumers
20 kcal/m²/yr
Heat
Secondary consumers
200 kcal/m²/yr
Heat
DECOMPOSERS
Primary consumers
2,000 kcal/m²/yr
Heat
Primary producers
20,000 kcal/m²/yr
Light energy
Energy stored as biomass

Biomagnification

- Certain environmental toxins can become more concentrated as they move up the food chain, with the highest concentrations occurring in the top consumers, a process called **biomagnification**.
- Essentially, a top consumer ingests all the toxins that had previously accumulated in the bodies of the organisms at the lower trophic levels.
- This explains why frequently eating certain fish, like tuna or swordfish, increases mercury exposure, a toxic heavy metal.


Mercury
Volcano
Mine
Coal plant
Methylmercury
FOOD CHAIN
Shark
Pike
Albacore
Halibut
Trout
Tuna
(can)
Krill
Salmon
Pollock
Oyster
Mercury level
(EPA advice for
consumption)
Eat only a few
times per month
Eat a few
times per week
Unlimited


PCBs are released from multiple sources into Puget Sound
1
PHYTOPLANKTON ABSORB PCBs
2
3
ZOOPLANKTON
EAT PHYTOPLANKTON
HERRING EAT
ZOOPLANKTON
4
SALMON
EAT HERRING
Biomagnification
LARGE MAMMALS
(ORCA WHALES)
EAT SALMON
Chart not to scale

Food Web

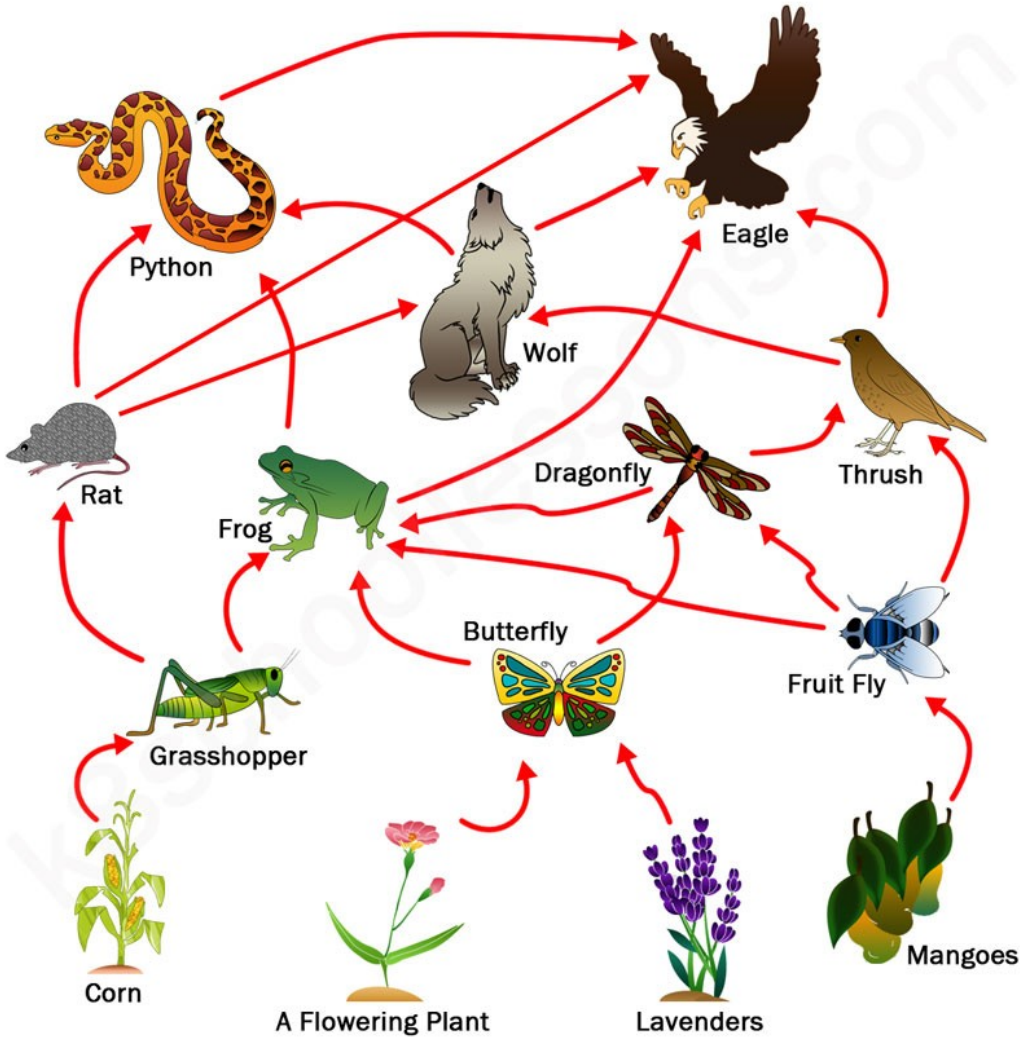
Food webs are networks of **interconnected food chains** that show how energy and nutrients move through an ecosystem.

- Consists of **multiple interconnected food chains**.
- Show that organisms often have **more than one food source**.
- Illustrate the **feeding relationships** among different organisms.
- Demonstrate how **energy flows** through an ecosystem.
- Help explain how changes in one species can affect many others.

Importance of Food Webs

- Show ecosystem complexity.
- Help maintain ecosystem stability.
- Reveal the impact of losing or adding a species.

A Food Web


Python
Eagle
Wolf
Rat
Frog
Dragonfly
Thrush
Butterfly
Fruit Fly
Grasshopper
Corn
A Flowering Plant
Lavenders
Mangoes

Ecosystem Concepts Summary

Ecosystem Concepts Summary

- **Flow of Energy** – Energy moves in one direction through an ecosystem: **Sun → Producers → Consumers → Decomposers**.
- **Food Chain** – A simple pathway showing how energy and nutrients pass from one organism to another through feeding relationships.
- **Trophic Level** – The feeding position an organism occupies in a food chain.
  - 1st: Producers
  - 2nd: Primary Consumers
  - 3rd: Secondary Consumers
  - 4th: Tertiary Consumers
- **10% Energy Rule** – Only about **10% of the energy** at one trophic level is transferred to the next; about **90% is lost** as heat and life processes.
- **Biomagnification** – The process by which harmful substances (e.g., pesticides, mercury) become **more concentrated at higher trophic levels** in a food chain.
- **Food Web** – A network of interconnected food chains showing multiple feeding relationships within an ecosystem.

- These concepts help us understand **how energy flows** through ecosystems, **how organisms depend on one another**, and **why biodiversity and environmental protection are important** for **maintaining a balanced and healthy ecosystem**

Activity Time!

Think - Pair - Share



Think

Think about the question you have been asked, what your answer may be, and how you will share your thoughts out loud.

This is done quietly in your head.



Pair

When the teacher says, you will pair up with a partner. Be sure to listen for if your teacher will be assigning you your partner or if you will choose your own,



Share

Take turns sharing your answers or thoughts with your partner. One partner goes first and the other onw listens and then you will switch.

Good partners make eye contact and keep a voice level 0 when listening to their partner share. Both partners must share.